Sarah Boyd

Lab partner: Edwin Velasquez

5/5/25

Three phase Induction Motors

Introduction:

Three-phase induction motors are some of the most commonly used motors in the world. They are popular and often chosen over DC motors because they are simpler, reliable, and require very little maintenance. Besides checking the bearings, there's limited maintenance needs- these motors don't use brushes like some other motors, so there are fewer parts that can wear out or break.

The way a three-phase induction motor works is based on a basic idea called electromagnetic induction. When three alternating currents (AC) flow through coils in the motor's stator (the outer part of the motor), they create a rotating magnetic field. This spinning magnetic field passes through the rotor (the inside part that turns), which makes a current in the rotor and causes it to spin. That spinning rotor is what creates useful motion, or torque, that can do work. More information about how a three phase motor works is at the end of the lab.

Three-phase motors are more efficient and powerful than single-phase motors because they create a smoother and stronger rotating magnetic field. In a three-phase system, there are typically six magnetic poles—two poles for each of the three phases. This setup helps the motor run more smoothly and handle bigger jobs. Six poles versus two poles is included at the end of this lab. In this lab, we will learn how a three-phase induction motor works, how to measure its input and output power, and how efficient it is at turning electricity into mechanical motion. Calculation and content resources are listed at the end of the lab.

Objectives

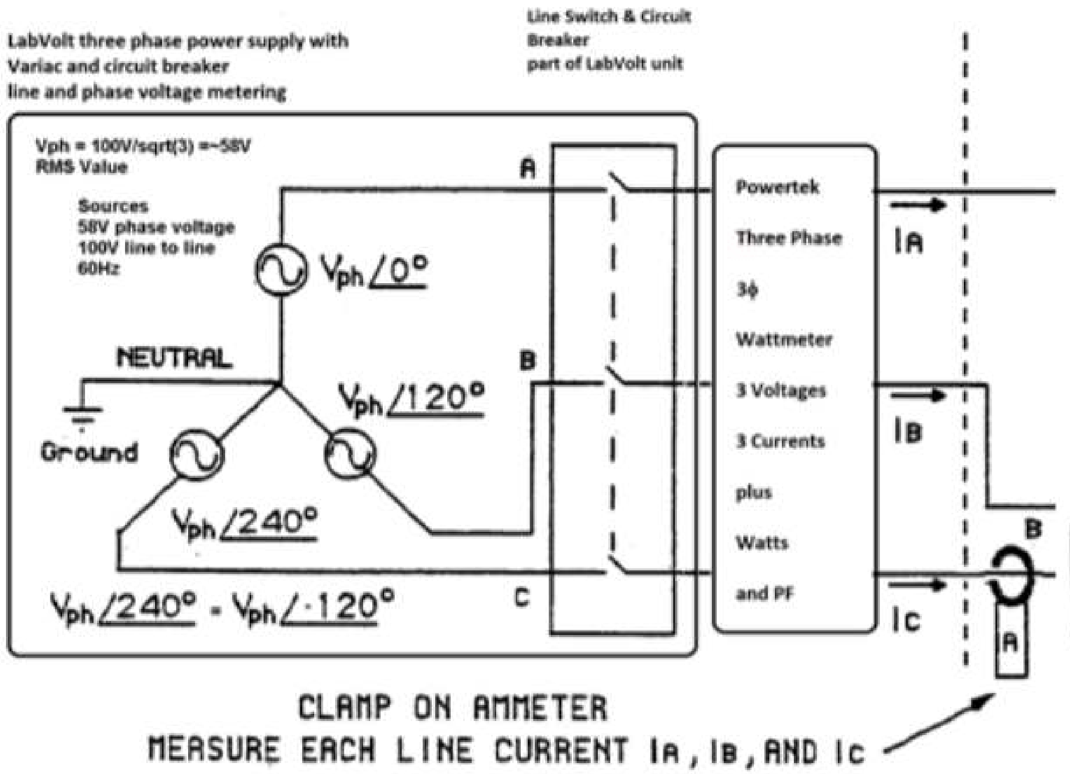
- Record mechanical output variables - torque and speed

- Electrical input variables Watts, VAR, VA and PF and use Pin (watts) to determine ACIM efficiency

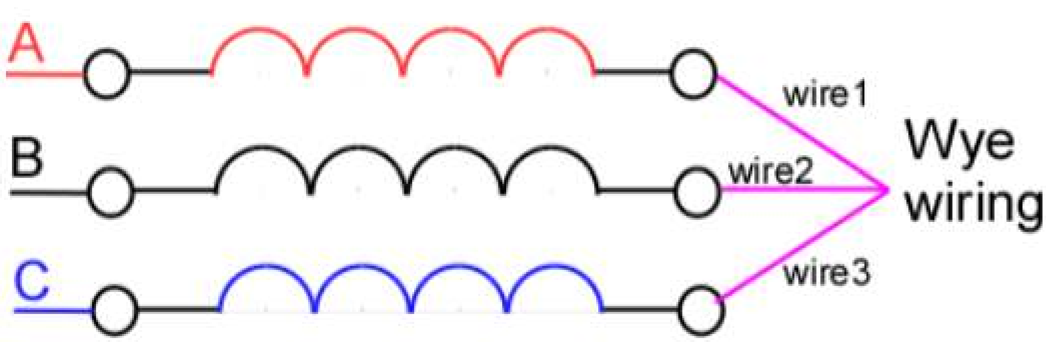
- To understand the typical connection and manual operation of an industrial variable speed/ frequency drive used with an AC induction motor

- Test a DC generator using a three phase motor as the prime mover or source of mechanical energy

Part 1



IMG 1. Schematic for physical circuit

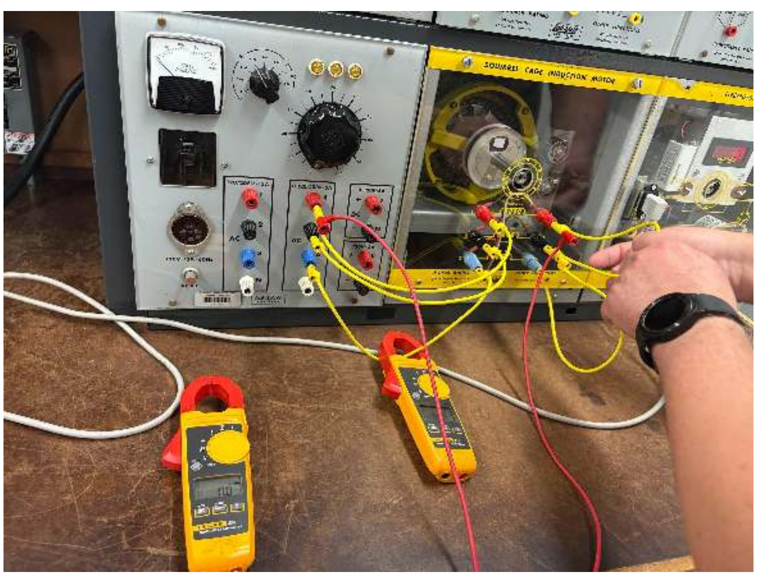


IMG 2. Wiring for physical circuit

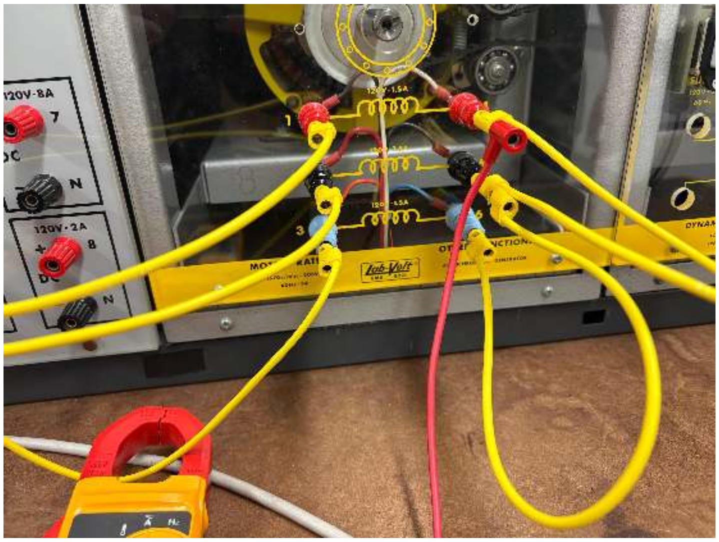
Voltage (volts)	Torque (N-m) ft/pound	Speed (RPM)	Armature current (Amps)
119.	0.6	1776	0.79
119.2	1	1775	0.8
119.2	2	1770	0.81
119.1	3	1763	0.84
119.0	4	1740	0.91
118.9	5	1730	0.96
118.8	6	1725	0.98
118.8	7	1715	1.03
118.7	8	1700	1.09
118.8	9	1680	1.18
118.7	10	1675	1.22

Table 1. *Data from changing the torque.*

As we add load, the torque changes and thus the speed and current changes.



IMG 3. *Circuit*



IMG 4. *Initial orientation of circuit*

6.One very useful feature of three phase motors is changing any two phases will reverse the direction of the motor by reversing the direct of the rotating magnetic field from one so from the standard rotation A-B-C-A-B-C if we switch A and B we get B-A-C-B-A-C or A-C-B rotation as does switching B and C phases A-C-B--A-C-B. This means we only need a DPDT to reverse the motor (A and C are often reversed C-B-A--C-B-A leaving B to be unchanged but that is not required by code (by our professor's research).

This aspect of the three-phase motor was confirmed and can be seen in this video here:

https://youtu.be/vn69FTSbHLo?si=KghHOBYqTua8c1YQ

Part 2:

Then, we lowered the frequency by 5 hz each time and recorded the results with the load switched on and the load off. We used three phase pwm to adjust voltage and frequency to drive the meter.

Voltage to the motor v	Frequency hz of the motor	Voltage of the generator (multimeter)	load	Speed (RM)
117	60	160.6	0	1800
115.7	60	117.7	56 ohms	1540
117.0	55	146.9	0	1600
115.9	55	111.4	56	1460
111.11	50	133.5	0	1450
110.8	50	103.1	56	1340
100.5	45	119.9	0	1320
100.3	45	93.2	56	1220
89.3	40	105.7	0	1190
84.3	40	82.6	56	1080
78.6	35	92.5	0	1020
67.8	30	79.3	0	900
56.6	25	65.7	0	720
45.9	20	52.1	0	565

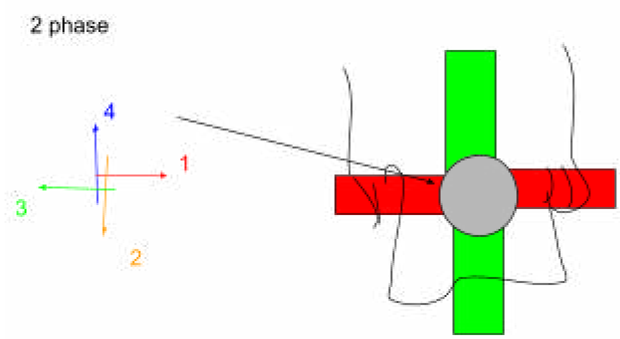
Table 2. *Data from changing the frequency.*

The motor's frequency is directly related to its speed, based on the formula speed (RPM) = (120 × Frequency in Hz) / Number of Poles. For example, with a 4-pole motor and a frequency of 72 Hz, the speed would be: Speed = (120 × 72) / 4 = 2160 RPM. We also used a stroboscopic light to measure motor speed. When the flashing rate of the light matches the motor's rotation speed, the pattern on the motor appears to stand still. This effect is called the stroboscopic effect, and it helps us visually confirm rotational speed without making contact.We controlled the motor using a combination of voltage and speed control through the PWM method. The direction of the motor was reversed electronically. Previously, we had to physically swap the motor wires, but in this lab we used a reversing button built into the control system.

Conclusion

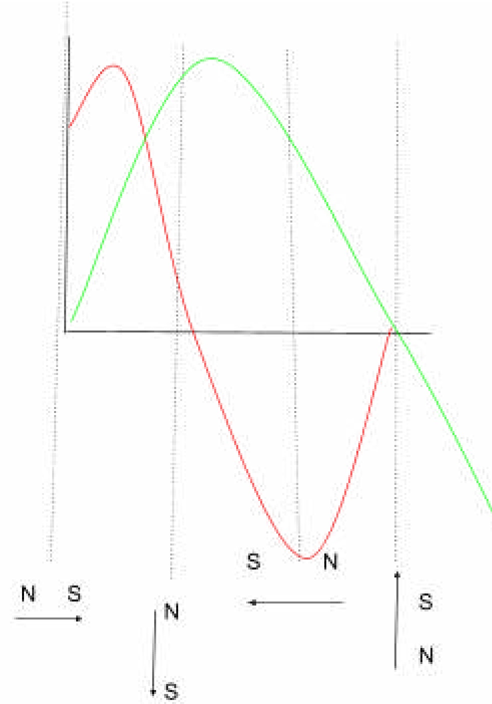
In this lab, we explored how the direction of a three-phase induction motor can be reversed by swapping any two of the three phase wires, which reverses the rotating magnetic field. This simple method allows for motor direction control using just a DPDT switch. We also experimented with changing the motor's frequency, lowering it by 5 Hz increments, and recorded the motor's behavior both with and without a load. Using three-phase PWM, we adjusted both voltage and frequency to control the motor's performance. The relationship between frequency and speed was confirmed by our data and the concepts behind the formula speed (RPM) =(120 f)/poles.We verified motor speed using a stroboscopic light, which made the rotor pattern appear still when the flash rate matched the motor’s rotation. Direction control was handled electronically using a built-in reversing function instead of manually swapping wires. Overall, the lab demonstrated key principles of speed control, direction reversal, and non-contact speed measurement in three-phase motor systems.

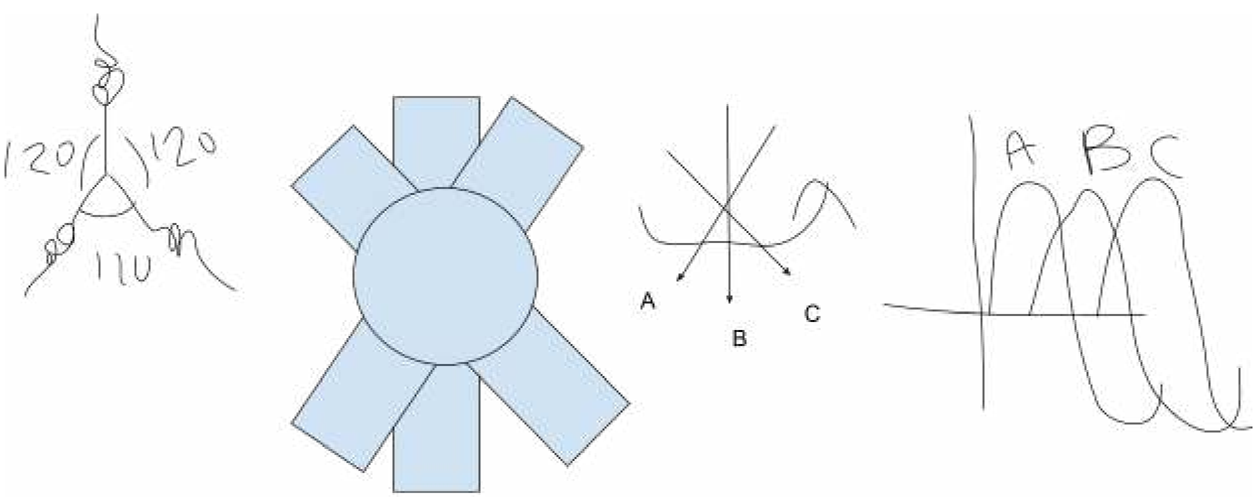
Resources



Creates alternating magnetic field

2 phase * 2 =4 poles

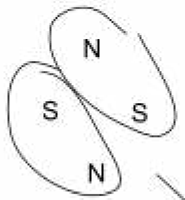




For three phase, its 6 poles. 2 poles per phase, so 2 poles *3 phases =6 poles.

$$\text{Synchronous speed} = \frac{120\ F}{\#P} = \frac{120\ (60hz)}{4} = 1800$$

1795rpm , how many poles?

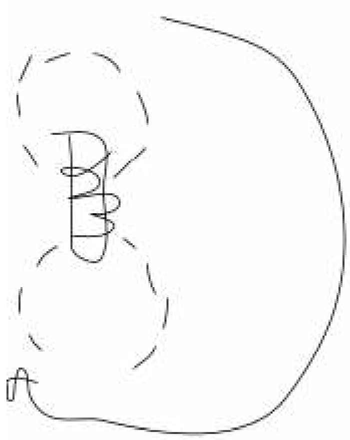


Given or measured w/ the dino

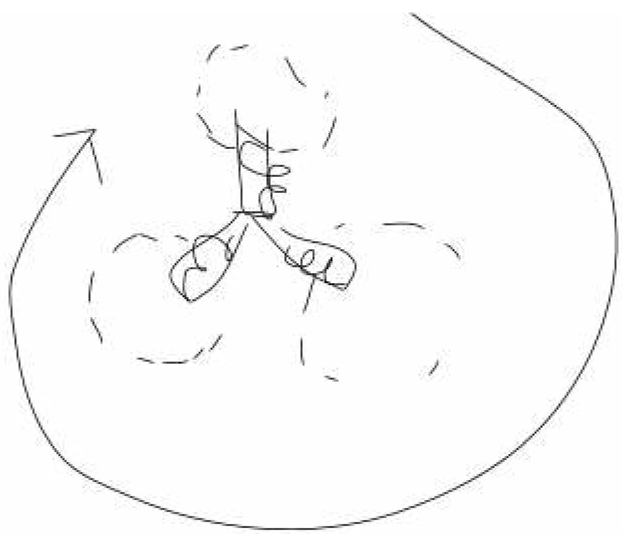
4 poles total, two pairs

$$\text{Slip speed} = \frac{(Nsync-Nrotor)}{Nsync} = \frac{(\text{syncronous speed- Speed of rotor})}{\text{Synchronous speed}} * 100$$

Slip =1, 0 speed, Slip =0, same speed, slip expressed in %



Two pole stalls where stator doesn't have permanent magnets



When there's three poles it is more efficient